Predicting How SJSU Students Commute: Analysis of Machine Learning Accuracy and Fairness

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Specifications:

1. **Task:** Classify a student's primary choice for commuting (driving, carpool, transit, walk, bike).
2. **Features:** Look at the distance of the commute, housing condition (perhaps neighborhoods as well), x amount of days commuting to campus, does student have a parking/transit pass, and enrollment status.
3. **Group attributes for fairness:** Age, gender, income or financial aid.
4. **Models:** Logistic regression/decision tree.
5. **Evaluation:** Look at accuracy, and retrain to reduce sensitive features then compare the accuracy and fairness tradeoff.
6. **Tools:** Python and pandas (more TBD)

Data Sources:

- **Primary Source:** The SJSU Annual Student Commute Survey, most recent release available. May contact SJSU Transportation Solutions and ask for an anonymized response-level for data for coursework. This is because the data that is published is almost entirely aggregated, meaning we are unable to train a classifier on that.
- **Backup:** National Household Travel Survey (NHTS) person-level data, filtered to students.

Report Outline:

1. Introduction
2. Related work
3. Dataset and exploration analysis
4. Preprocessing/methods
5. Results
6. Conclusion

Roles:

- Brian Wallichs : Data collection and cleaning; will look at fairness analysis.
- Seline Venzon: Model training and eval; experiments pertaining to accuracy vs. fairness.
- Both students will contribute to the research and documentation of this project.

Repository:
https://github.com/bwallichs/CMPE-108-Project